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Electronic modulation of Flower-petal-shaped Ni–Co–P nanoarray doping by Fe and Mo for efficient overall water splitting

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ABSTRACT

In light of the promise of electrolytic water-to-hydrogen technology, it is urgent to develop bifunctional catalysts capable of simultaneous hydrogen/oxygen evolution reactions (HER/OER). This paper prepared a self-supported NiCoFeMoP/NF catalyst, which constructed Ni–Co–P nanoarray on nickel foam and modulated the electronic structure by doping Fe (2.0 wt%) and Mo (6.9 wt%). The results indicated that, in the hydrogen evolution reaction, the overpotential of the NiCoFeMoP/NF catalyst at a current density of −100 mA cm^{-2} was 146 mV, which was lower than that of the Pt/C electrode. In the oxygen evolution reaction, the overpotential of the NiCoFeMoP/NF catalyst at a current density of 50 mA cm^{-2} was 260 mV, which was lower than that of the RuO_2 electrode. In the overall water splitting, the NiCoFeMoP/NF catalyst required only 1.558 V to reach a current density of 10 mA cm^{-2} and exhibited exceptional stability. Experimental and computational results demonstrate that the electronic structure of Ni–Co–P can be modulated by incorporating Fe and Mo elements. This modification optimizes the Gibbs free energy of HER/OER intermediates, expedites the kinetic process of water splitting, and enhances the catalytic activity.

1. Introduction

Efficient electrocatalysts can significantly reduce the overpotentials caused by the slow transfer of multiple proton-coupled electrons in the hydrogen evolution reaction (HER) and oxygen evolution reaction (OER) [1,2]. The most effective electrocatalysts for HER and OER are still Pt-based [3] and Ru/Ir-based [4,5]materials. However, insufficient reserves, high costs, and short service life limit the scale-up of these catalysts. Additionally, the incompatibility and mechanisms of OER and HER may lead to inefficiencies [6]. Therefore, designing efficient, economical, and stable bifunctional integral water separation catalysts is imminent. To date, carbon-based materials [7–9] and non-precious metal materials such as transition metal compounds (including sulfides [10], oxides [11], nitrides [12], and phosphides) have been widely used as electrocatalysts for HER and OER [13,14]. Transition metal phosphides (TMPs) have been extensively studied due to their multiple electronic orbitals, metallic properties, and low cost [15,16]. It has been reported that in the HER process, the phosphorus site (P-site) and the metal site (M-site) can be active sites. The highly electronegative P-site attracts protons from the solution, promoting the formation of metal hydrides and facilitating the release of H_2 molecules [17]. For example, Sun et al. [18] prepared CoP-NC/PET electrodes by anchoring amorphous CoP catalysts on flexible fabrics, which exhibited excellent HER performance in alkaline environments with an overpotential of only 63 mV at a current density of 10 mA cm^{-2}.

However, compared to Pt/C or RuO_2/IrO_2 electrodes, single-metal TMP still exhibits a significant gap in electrochemical performance and structural stability. This disparity may be attributed to insufficient exposure of active sites, poor electrical conductivity, the susceptibility of supporting catalysts to corrosion by electrolytes, and the tendency for active components to decompose [19,20]. In contrast to monometallic phosphides, bimetallic phosphides such as NiCoP [21,22], CoMoP [23,24], and NiFeP [25,26] exhibit higher electrical conductivity and synergistic effects among their components, resulting in higher electrocatalytic activity. Guo et al. [27]. Developed a novel self-supported NiCoP electrode on nickel foam, achieving a HER overpotential of 57

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mV and an OER overpotential of only 256 mV at a current density of 10 mA cm^{-2}. Its OER performance surpassed that of RuO_2 and IrO_2. Although some remarkable progress has been made, the performance of single TMPs is still unsatisfactory. This is primarily attributed to the overly strong interaction between phosphorus (P) and adsorbed hydrogen (H_{ads}), which often hinders the desorption of H_{ads} to form H_2. As a result, their catalytic performance and stability still fall short of meeting the demands of industrial applications [28]. Studies in recent years have shown that heteroatom doping engineering is easy to implement and can modulate the electronic structure, reduce the reaction energy barrier, accelerate the reaction kinetics, and fundamentally enhance the intrinsic activity [29–31]. Ramadoss et al. [32]. Report a self-template representation of self-assembled iron-modulated cobalt-nickel phosphide (Fe–CoNiP) grown on 3D-nickel foam (NF) using a facile amine hydrolysis-approached synergetic phosphorylation strategy, these Fe–CoNiP nanoarchitectures exhibit excellent electrocatalytic hydrogen and oxygen evolution reactions with very low applied overpotentials of 110 and 280 mV to obtain 10 mA cm^{-2}, respectively.

While the HER and OER performance of the catalyst is improved by doping these metal atoms with bimetallic phosphide, they have not achieved the activity of precious metals in alkaline dielectrics, so it is necessary to choose more suitable metal atoms for doping [33]. To date, different d-orbital metals (e.g., Mo [34], Cu [35], Fe [36], Mn [37], etc.) are doped into TMP to enhance its hydrolysis dissociation activity. Among these dopants, 3d-metal Fe and non-3d-metal Mo are both low in cost and abundant and have different metal orbitals. This difference allows them to adjust the electronic structure compared to a single dopant effectively, optimizing the adsorption-free energies of the HER/OER reaction intermediates and accelerating the hydrolysis dissociation kinetic process [38].

This paper prepared a NiCoFeMoP/NF bifunctional electrocatalyst using a one-step hydrothermal method and a low-temperature phosphitylation method for overall water splitting. We innovatively doped Fe and Mo simultaneously onto NiCoP nanoarrays to modulate their electronic structure and microscopic morphology, thereby enhancing electrolytic water performance. In the hydrogen evolution reaction, NiCoFeMoP/NF catalyst overpotentials were 58, 117, and 146 mV at −10, −50, and −100 mA cm^{-2} current densities, respectively. The overpotentials of Pt/C catalyst at the same current densities were 29, 109, and 185 mV. The HER performance of the NiCoFeMoP/NF catalyst gradually surpassed that of the Pt/C catalyst as the current density increased. NiCoFeMoP/NF catalysts exhibit outstanding HER electrocatalytic activity compared to most other materials. In the oxygen evolution reaction, the overpotentials of the NiCoFeMoP/NF catalyst at current densities of 10, 20, and 50 mA cm^{-2} were 236 mV, 245 mV, and 260 mV, respectively. The OER performance of this catalyst is superior to that of NiCoFeP/NF and MoP/NF and even to that of RuO_2, exhibiting excellent OER electrocatalytic activity. In the overall water-splitting process, the NiCoFeMoP/NF catalyst required only 1.558 V to achieve a current density of 10 mA cm^{-2} and exhibited exceptional stability. Experimental and density functional theory (DFT) calculations show that Fe and Mo doping modulate the electronic structure, increase the intrinsic catalytic activity of the catalysts, and optimize the adsorption energies of hydrogen and oxygen intermediates (H*, OH*, and OOH*) on the surface of the catalysts. This optimization resulted in the improvement of HER/OER kinetic processes.

2. Experimental sections

2.1. Chemicals and reagents

Cobalt nitrate hexahydrate (Co $(NO_3)_2{\cdot}6H_2O$), Ferric nitrate hydrate (Fe $(NO_3)_3{\cdot}9H_2O$), Nickel nitrate hexahydrate (Ni $(NO_3)_2{\cdot}6H_2O$), Ammonium fluoride (NH_4F), Urea (CH_4N_2O), Sodium molybdate ($Na_2MoO_4{\cdot}2H_2O$), Sodium Hypophosphite ($NaH_2PO_2{\cdot}H_2O$) and Nitric acid (HNO_3) are analytical grade.

2.2. Synthesis of NiCoFeMoP/NF

In the hydrothermal synthesis experiments, $Co(NO_3)_2{\cdot}6H_2O$ (1 mmol), Fe $(NO_3)_3{\cdot}9H_2O$ (0.25 mmol), $Ni(NO_3)_2{\cdot}6H_2O$ (0.5 mmol), NH_4F (3.5 mmol), CH_4N_2O (6.5 mmol), $Na_2MoO_4{\cdot}2H_2O$ (1 mmol), and HNO_3 (66 μL) were successively dissolved in 25 mL of deionized water. The resulting solution and the pretreated nickel foam were transferred to a 50 mL hydrothermal synthesis reactor and heated at 120 °C for 6 h. Remove the nickel foam with the grown material after the reactor naturally cools to room temperature. It was rinsed several times with deionized water and anhydrous ethanol, then vacuum-dried in a vacuum-drying oven at 60 °C to obtain the NiCoFeMo/NF material. In the gas flux, a porcelain boat containing $NaH_2PO_2{\cdot}H_2O$ (1 g) was placed upstream of the tube furnace. In contrast, a porcelain boat containing NiCoFeMo/NF was placed downstream of the gas in the tube furnace for the low-temperature phosphating reaction. Nitrogen continuously flowed in as a protective gas. The tube furnace was heated to 400 °C at a heating rate of 2 °C min^{-1} and maintained for 2 h. After cooling to room temperature, the synthesized self-supported electrodes were removed, rinsed several times with deionized water and anhydrous ethanol, and dried in a vacuum drying oven at 60 °C to obtain the NiCoFeMoP/NF electrode.

2.3. Synthesis of NiCoFeP/NF

In the experiment, Co $(NO_3)_2{\cdot}6H_2O$ (1 mmol), Fe $(NO_3)_3{\cdot}9H_2O$ (0.25 mmol), Ni $(NO_3)_2{\cdot}6H_2O$ (0.5 mmol), NH_4F (3.5 mmol), and CH_4N_2O (6.5 mmol) were sequentially dissolved in 25 mL of deionized water under magnetic stirring for hydrothermal synthesis. The NiCoFeP/NF electrode was finally obtained through a low-temperature phosphorylation reaction.

2.4. Synthesis of MoP/NF

NH_4F (3.5 mmol), CH_4N_2O (6.5 mmol), and $Na_2MoO_4{\cdot}2H_2O$ (1 mmol) were sequentially dissolved in 25 mL of deionized water under magnetic stirring for hydrothermal synthesis. The MoP/NF electrode was finally obtained through a low-temperature phosphorylation reaction.

2.5. Preparation of Pt/C and RuO2 electrodes

20 mg of 40 wt% Pt/C (or RuO_2) was dispersed into a mixture of 180 μL pure water, 20 μL of Nafion, and 200 μL of anhydrous ethanol. The solution was then ultrasonicated for 1 h to form a homogeneous catalyst mixture. Finally, 40 μL of the catalyst ink mentioned above was uniformly dripped onto the nickel foam and vacuum-dried in a vacuum drying oven at 60 °C to obtain a Pt/C (or RuO_2) electrode with a loading mass of 2 mg $cm^{-2.}$

2.6. Material characterization

The X-ray diffraction patterns (XRD) of the prepared samples are detected from a Smart Lab X-ray diffractometer, which uses Cu Kα radiation as an X-ray source (Rigaku Corporation, Japan, λ = 1.5418 Å), and the measurement range of 2θ is 10°–80°. The microstructure analysis is explored by Merlin Compact scanning electron microscope (Carl Zeiss NTS GmbH, Germany). The transmission electron microscopy (TEM) images and elementary composition of the experimental samples are detected using a JEOL JEM 2100F (Japan) high-resolution transmission electron microscope, which is armed with energy dispersive spectroscopy (EDS). The Thermo ESCALAB 250XI (Thermo Scientific K-Alpha, America) X-ray photoelectron spectroscopy (XPS) with Al Kα (1486.6Ev) X-ray source is carried out to ascertain the chemical states of

elements on the surface of the sample.

2.7. Electrochemical measurements

All electrochemical measurements were conducted using an electrochemical workstation (CHI760E) in a three-electrode system. A prepared self-supporting electrode was employed as the working electrode, a graphite rod served as the auxiliary electrode, a saturated calomel electrode (SCE) was used as the reference electrode, and a 1.0 M KOH solution was utilized as the electrolyte. Before conducting electrochemical measurements of the hydrogen evolution reaction (or oxygen evolution reaction), the electrolyte was saturated with nitrogen (or oxygen) by bubbling nitrogen (or oxygen) through the electrolyte for 30 min. The working electrode was immersed in a KOH solution with a geometric area of 1 cm × 1 cm. The potential of the saturated calomel electrode was converted to the potential of the reversible hydrogen electrode (RHE) using the equation ERHE = ESCE + 0.059 × pH + 0.242. LSV polarization and Tafel plots were obtained at a scan rate of 2 mV s^{-1} under IR compensation conditions. The magnitude of the solution resistance (Rs) and charge transfer resistance (Rct) of the catalytic electrode during the reaction was measured using the AC impedance method in the frequency range of 10^5–0.01 with the voltage applied for the hydrogen evolution of −1.26 V (vs. and for and the oxygen evolution was of 0.46 V (vs. SCE), with an amplitude of 0.005 V. The voltage applied to the catalytic electrode was 0.005 V for the hydrogen evolution reaction. The electrochemically active specific surface area (ECSA) was measured in the non-Faraday potential region at different scan rates (10, 20, 40, 60, 80, and 100 mV s^{-1}). The hydrogen evolution reaction, oxygen evolution reaction, and overall water splitting stability tests were performed using three different methods: a current density of 10 mA cm^{-2} for a 100-h chronopotentiostatic test at a current density of 10–200 mA cm^{-2}, a short-term chronopotentiostatic test at a stepped current density of 10–200 mA cm^{-2}, and linear sweep voltammetry (LSV) curves after 100 h of stability versus 1000 cycles of cyclic voltammetry (CV) test.

2.8. Density functional theory (DFT) calculations

DFT calculations were performed in the Vienna ab initio simulation package (VASP) [39], and the results were post-processed using the Vaspkit [40] program package. In the structure optimization and self-consistent calculations, the electronic interactions are described using the GGA-PBE (Perdew-Burke-Ernzerh of generalized gradient exchange approximation) [41] exchange-correlation generalization with the spin polarization turned on, and the kinetic energy truncation energy is set to 450 eV for all plane-wave basis sets. Based on the high-resolution transmission electron microscopy (HRTEM) images and XRD analysis, we selected NiCoP (111) as the catalytic surface for the DFT calculation. The Brillouin zone was delineated using 3 × 3 × 1 Monkhorst-Pack grids for the NiCoP (111), NiCoFeP (111), and NiCoFeMoP (111) and optimized using the CG (Conjugate gradient) algorithm with van der Waals corrections by the DFT-D3 [42] method, the convergence of the energy and force parameters were set to 1 × 10^{-6} and 0.02 eV A^{-1}. In order to take into account the effect of temperature, the zero-point energy at 298 K is corrected in the calculation of all free energies.

3. Results and discussion

3.1. Structural characterization

NiCoFeMoP/NF catalytic electrodes were prepared using hydrothermal and low-temperature phosphitylation according to the synthesis route shown in Fig. 1.

The morphology and structure of NiCoFeMoP/NF were characterized (Fig. 2). Fig. 2a shows the structural and compositional information by

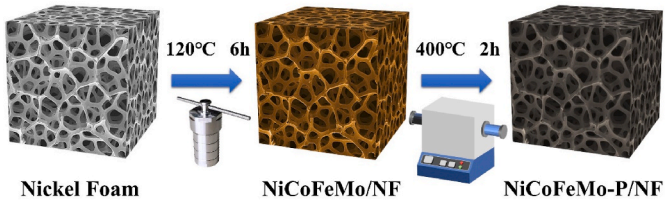


Fig. 1. Schematic diagram of the synthesis of NiCoFeMoP/NF materials.

X-ray diffraction (XRD), the three high-intensity diffraction peaks at 44.507°, 51.846°, and 76.370° are attributed to the nickel foam substrate. However, due to the strong diffraction peaks of the nickel foam substrate, the intensity of the diffraction peaks of the material loaded on the nickel foam is significantly reduced. Upon logarithmic transformation of the XRD pattern, the diffraction peaks at 26.578° and 40.987° correspond to the NiCoP (PDF#71–2336) material. After logging the XRD patterns, the diffraction peaks at positions such as 26.578° and 40.987° correspond to the NiCoP (PDF#71–2336) material, indicating that the NiCoP material was successfully deposited on the nickel foam.

The morphology of nickel foam and NiCoFeMoP/NF was observed using a transmission electron microscope (TEM) and scanning electron microscopy (SEM).

Fig. 2b TEM image shows a flower-like cluster morphology with lamellar structures crossing each other, consistent with the morphology observed by SEM of NiCoFeMoP/NF. From Fig. 2c HRTEM image, the crystal plane spacing of the NiCoFeMoP/NF electrode material is 0.22 nm. The lattice stripe corresponds to the (111) crystal plane of NiCoP, and the diffraction peak located at 40.987° in the XRD also corresponds to the (111) crystal plane. The TEM and the XRD crystal planes are consistent, confirming that the synthesized catalyst is Ni–Co–P.

Fig. 2d–f shows the SEM images of NiCoFeMoP/NF catalysts at different magnifications, respectively. Fig. S1a shows the SEM image of nickel foam. It can be observed that the NiCoFeMoP material completely covers the nickel foam, exhibiting a three-dimensional structure. This structure provides a larger surface area for the electrocatalyst, facilitating the exposure of more catalytic active sites to the electrolyte and accelerating gas desorption [43]. Additionally, the material initially forms a layer of nanosheets on the nickel foam, followed by the growth of nanosheet-like clusters on this layer of nanosheets, forming sheet flower-like clusters.

The elemental composition and surface distribution of the NiCoFeMoP/NF catalysts were analyzed using EDS elemental analysis. Fig. 2g and h clearly shows the distribution of Ni (orange), Co (blue), Fe (yellow), Mo (purple), P (green), and O (red) elements on the surface of the lamellar nanostructures. Fig. 2i–k demonstrates the uniform distribution of Ni, Co, and Fe elements on the NiCoFeMoP/NF electrocatalysts. XRD and TEM analysis confirmed that the synthesized substance was Ni–Co–P, with uniform Fe elements on the surface. It was speculated that Fe elements were incorporated into the NiCoP substance, while Mo, P, and O elements were located in the outermost region of the 3D nanostructured nanoclusters (Fig. 2l–n). The elemental composition indicates that the MoP substance could have formed during the low-temperature phosphitylation process. The results suggest that NiCoFeMoP/NF nanoclusters were produced during the low-temperature phosphitylation of NiCoFeMoP/NF. The above results demonstrate the preparation of NiCoFeMoP/NF.

Further investigation was conducted by XPS on the composition and valence states of the NiCoFeMoP/NF electrocatalysts. Fig. 3 is the XPS characterization. The spectra of the Ni 2p electrocatalysts are shown in Fig. 3a. The high-resolution Ni 2p spectrum in Fig. 3b is deconvoluted into three spin-orbit double peaks, showing two typical peaks at binding energies of 853.47 and 870.61 eV, attributable to $2p_{3/2}$ and $2p_{1/2}$ of the $Ni^{\xi+}$ species bonded to P ($0 < \xi < 2$) in NiCoFeMoP/NF, respectively.

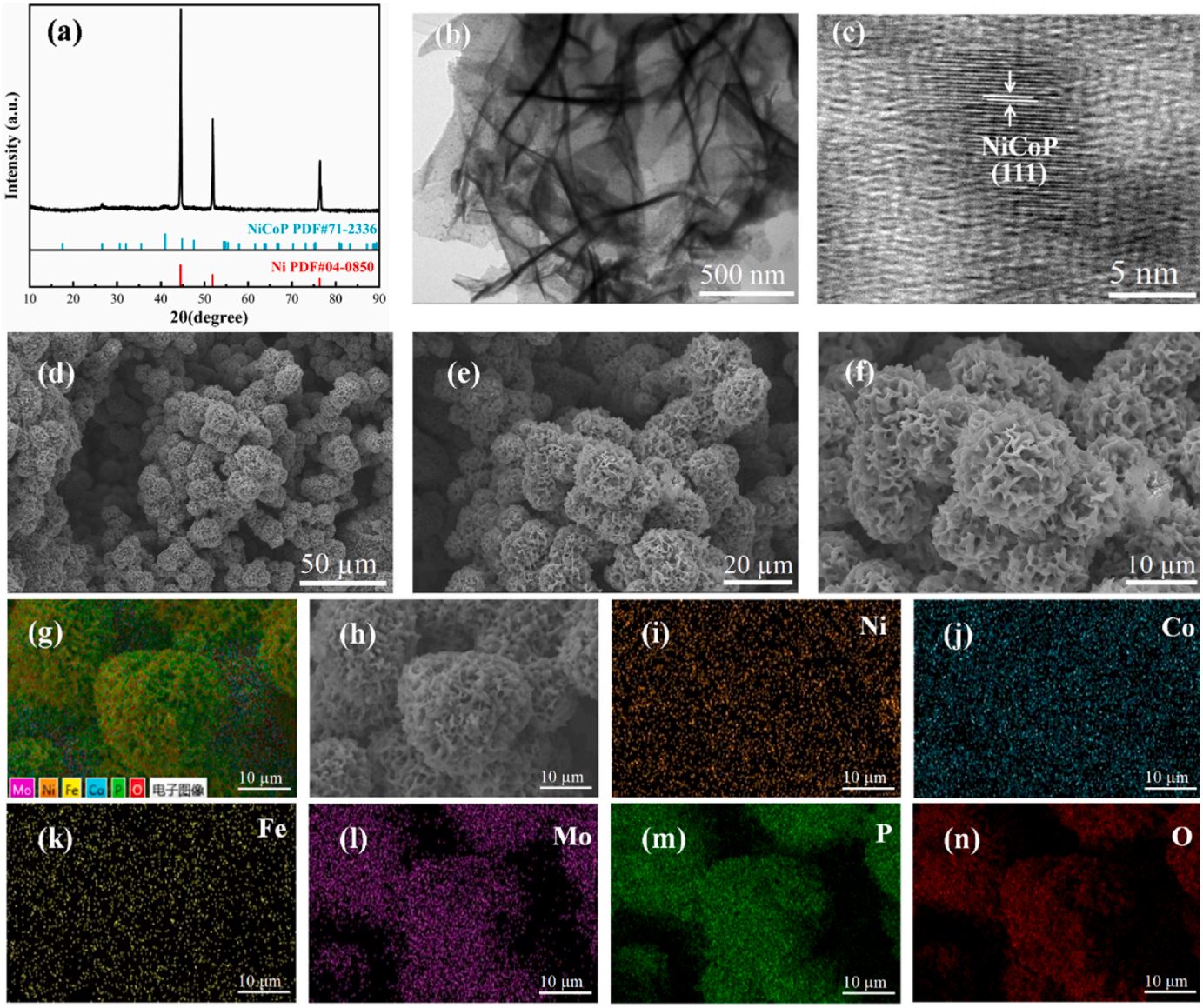


Fig. 2. Morphological and structural characterization of NiCoFeP. (a) XRD pattern, (b) TEM pattern, (c) HRTEM pattern, (d–f) SEM images of NiCoFeMoP/NF electrodes at different magnifications, (g–n) EDS-SEM pattern of NiCoFeMoP/NF electrode.

The peaks at 857.03 and 874.44 eV can be attributed to $2p_{3/2}$ and $2p_{1/2}$ of Ni^{2+} species. The binding energies around 861.32 and 879.77 eV in the Ni 2p spectrum are attributed to the vibrational satellite peaks of $2p_{3/2}$ and $2p_{1/2}$, respectively [44]. The high-resolution Co 2p spectrum in Fig. 3c is deconvoluted into three spin-orbit double peaks. Two peaks appear at binding energies of 778.73 and 793.47 eV, corresponding to $2p_{3/2}$ and $2p_{1/2}$, attributed to P-bonded $Co^{\&+}$ species ($0 < \& < 2$) in NiCoFeMoP/NF, respectively. Additionally, two peaks at 782.16 and 797.95 eV can be assigned to $2p_{3/2}$ and $2p_{1/2}$ of Co^{2+} species. The binding energies near 786.36 and 802.78 eV in the Co 2p spectrum are attributed to vibrational satellite peaks [44].

The high-resolution Fe 2p spectra in Fig. 3d show two peaks at binding energies of 710.84 and 723.64 eV, attributable to $2p_{3/2}$ and $2p_{1/2}$ of the Fe^{2+} species in NiCoFeMoP/NF, respectively. Additionally, two peaks at 713.32 and 727.07 eV can be attributed to $2p_{3/2}$ and $2p_{1/2}$ of the Fe^{3+} material. The Fe binding energies near 715.80 and 719.31 eV in the 2p spectrum are attributed to Fe^{2+} and Fe^{3+} vibrational satellite peaks, respectively [45–47].

The Mo 3d high-resolution spectra of NiCoFeMoP/NF (Fig. 3e) show that the peaks at binding energies of 230.55 and 233.63 eV belong to Mo^{4+} $3d_{5/2}$ and Mo^{4+} $3d_{3/2}$ of NiCoFeMoP/NF, and the Mo^{6+} signals at binding energies of 232.47 and 235.82 eV are probably formed by the exposure of Mo^{4+} to air oxidation formation. In addition, the $Mo^{\delta+}$ species ($0 < \delta < 4$) signal at a binding energy of 228.21 eV is a typical peak for MoP [48–50].

The P 2p spectra of NiCoFeMoP/NF in Fig. 3f indicate that the peaks centered on the binding energies at 129.71 and 130.61 eV correspond to $2p_{3/2}$ and $2p_{1/2}$ of P. The peak at 134.26 eV is attributed to the P–O bonding, possibly due to oxidatively generated PO_3^{-4} or P_2O_5 [49,51]. The above results indicate the successful synthesis of the NiCoFeMoP/NF catalyst.

3.2. HER performance

Fig. 4 is the measurement of electrochemical performance. Fig. 4a shows the LSV curves of the hydrogen evolution reaction of Pt/C, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF. The nickel foam exhibits the poorest hydrogen evolution performance, significantly lagging behind the other four catalysts. Additionally, the Pt/C electrode demonstrates excellent catalytic performance at low current densities.

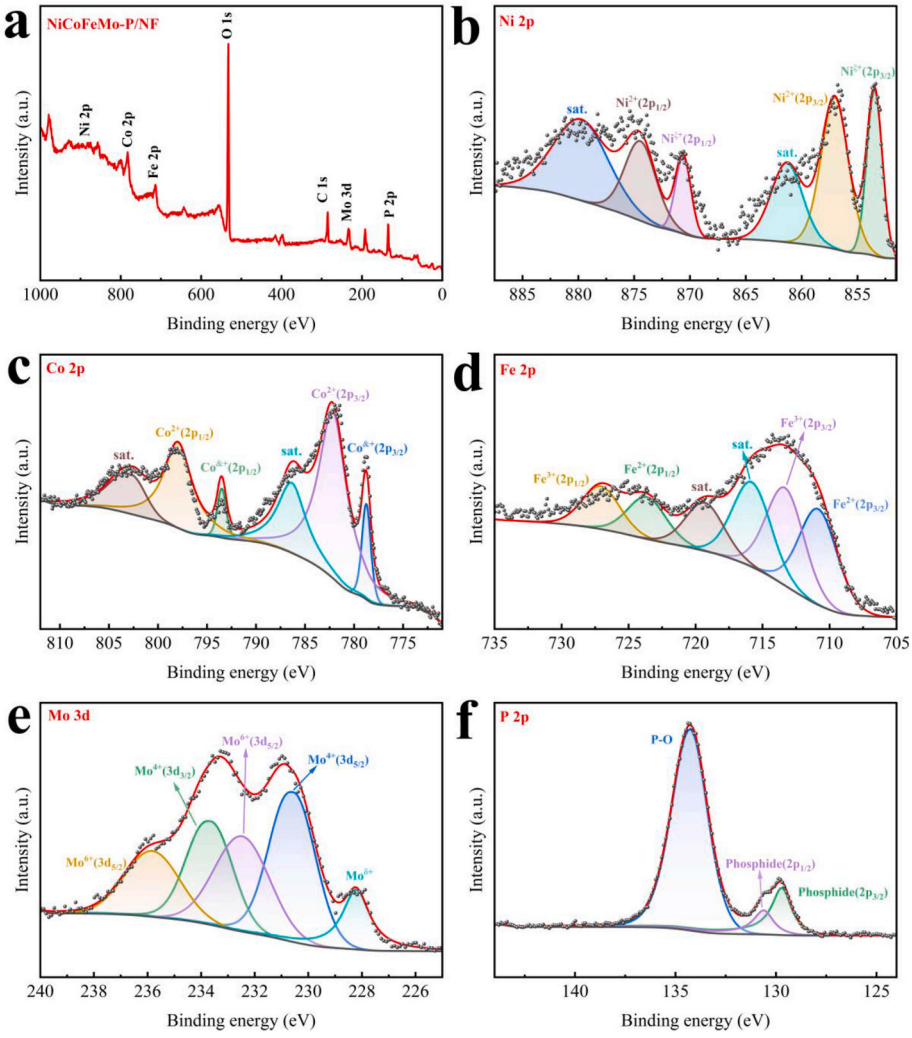


Fig. 3. Full XPS spectrum of (a) NiCoFeMoP/NF electrode, (b) Ni 2p, (c) Co 2p, (d) Fe 2p, (e) Mo 3d, (f) P 2p.

However, as the current density increases, the Pt/C electrode's performance gradually deteriorates compared to the NiCoFeMoP/NF catalytic electrode. The NiCoFeMoP/NF catalytic electrode exhibits superior hydrogen evolution performance compared to the NiCoFeP/NF and MoP/NF catalytic electrodes at high current densities. The bar charts of overpotentials at different current densities (Fig. 4b) show that the NiCoFeMoP/NF catalysts exhibit poorer hydrogen evolution performance than the Pt/C electrode at a current density of −10 mA cm^{-2}. However, at −50 mA cm^{-2}, the hydrogen evolution performance is essentially equal to that of the Pt/C catalyst. Moreover, at −100 mA cm^{-2}, the performance is better than that of the Pt/C electrode. Notably, the hydrogen evolution performance of the NiCoFeMoP/NF catalysts surpasses that of the NiCoFeP/NF, MoP/NF, and NF materials, leading to the highest HER electrocatalytic activity. The HER activity of the NiCoFeMoP/NF catalytic electrode was also significantly superior to that of most of the cobalt or nickel phosphides reported (Table 1).

The magnitude of the Tafel slope can reflect the fast and slow reaction kinetics of HER at catalytic electrodes. We calculated the Tafel slopes of the samples by transforming the IR-compensated LSV curves to assess the reaction kinetics in the HER process. Among the five catalysts, NiCoFeMoP/NF exhibited the most minor Tafel slope at 85 mV dec-1 (Fig. 4c). This result indicates that the NiCoFeMoP/NF catalytic electrode has faster reaction kinetics in the HER process. In addition, both the Tafel slope and the overpotential results obtained from the LSV curves demonstrated the excellent catalytic performance of the NiCoFeMoP/NF material. To further investigate the HER activity of Pt/C, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF, electrochemical impedance spectroscopy (EIS) tests were performed at −1.26 V (vs. SCE) (Fig. 4d).

Comparison of the electrochemical impedance diagrams of Pt/C, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF shows that the Rct of nickel foam is the largest, followed by the MoP/NF electrode. This indicates that nickel foam exhibits the slowest charge transfer efficiency and the poorest hydrogen evolution performance in the hydrogen evolution reaction. Moreover, the charge transfer resistance decreases, and the hydrogen evolution performance improves when MoP catalysts are applied to nickel foam. From the magnified detail of the electrochemical impedance (Fig. 4e), the NiCoFeMoP/NF catalyst exhibits the smallest Rct and the highest charge transfer efficiency. The high-frequency intersection of the curve with the X-axis in the electrochemical impedance spectrum represents Rs, the resistance primarily caused by the electrolyte and the exposed catalyst substance. At the same time, the diameter of the semicircle indicates Rct. To separate the ohmic resistance from the polarization resistance, we employ the model of Rs in series with the module, where Rct is connected in parallel with the constant phase element (CPE). The circuit fitting above results in Rcts (Fig. 4d and e) of 1.437, 0.727, 1.087, 3.993, and 11.97 Ω for Pt/C, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF, respectively. The NiCoFeMoP/NF catalyst has the smallest Rct of 0.727 Ω, indicating excellent electrical conductivity and superior electrocatalytic

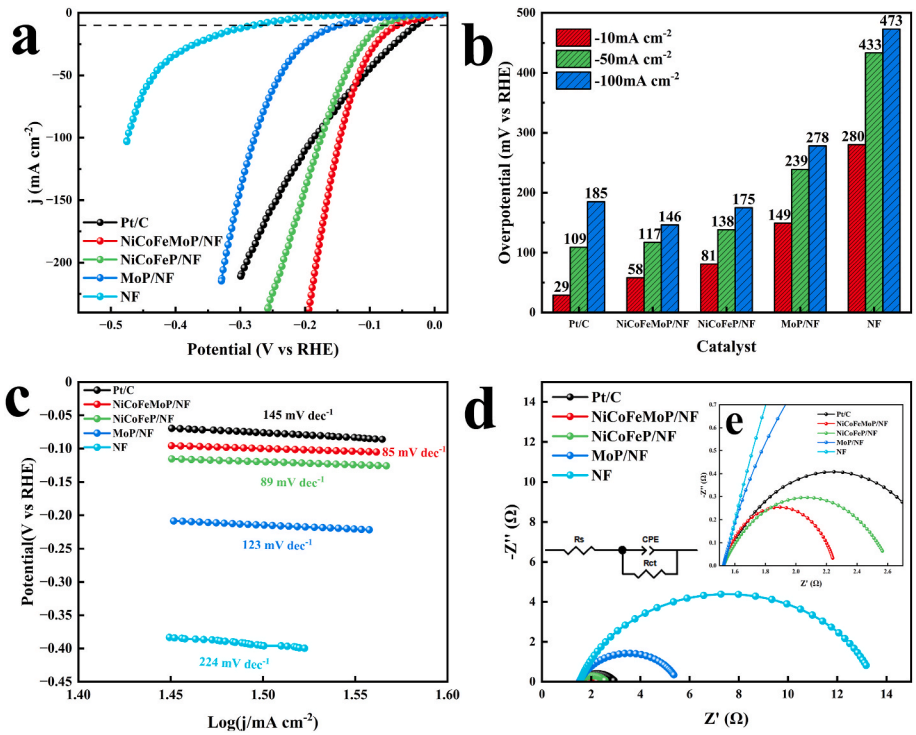


Fig. 4. Hydrogen evolution reaction of Pt/C, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF and NF materials with (a) LSV curves, (b) overpotential bar-contrast plots at different current densities, (c) Tafel plots, (d) electrochemical impedance spectra, and (e) zoomed in details of electrochemical impedance spectra.

Table 1
The comparison of the HER performance of the samples.

Sample	Overpotential (mV)	Stability	Electrolyte	Ref.
NiCoP	$\eta_{10} = 60$	24h	1 M KOH	[52]
Mo-doped CoP	$\eta_{10} = 118$	10h	1 M KOH	[53]
Fe-doped NiCoP	$\eta_{10} = 110$	48h	1 M KOH	[32]
Fe–CoP/NF	$\eta_{10} = 78$	10h	1 M KOH	[54]
CoMoP	$\eta_{10} = 126$	10h	1 M KOH	[55]
Ni, Fe-LDH@NCP/NF	$\eta_{10} = 102$	40h	1 M KOH	[56]
NiFeP@MXene	$\eta_{10} = 122$	12h	1 M KOH	[57]
V–CoP/CC	$\eta_{10} = 71$	25h	1 M KOH	[58]
Ni–CoP/HPFs	$\eta_{10} = 72$	21h	1 M KOH	[59]
CoP/NPC/TF	$\eta_{10} = 80$	10h	1 M KOH	[60]
NiCoFeMoP	$\eta_{10} = 58$	100h	1 M KOH	This work

performance in the alkaline hydrogen evolution reaction. This characteristic significantly enhances the charge transfer rate during the hydrogen evolution reaction.

3.3. OER performance

From the LSV plots of the oxygen evolution reactions of RuO_2, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF (Fig. 5a), the order of oxygen evolution performances is as follows: NiCoFeMoP/NF > NiCoFeP/NF > RuO_2 > MoP/NF > NF. The oxygen evolution performance of the NiCoFeMoP/N catalyst is superior to that of the other four catalysts. According to the bar charts depicting overpotentials at various current densities of the electrodes (Fig. 5b), the OER performance of the NiCoFeMoP/NF catalyst surpassed that of three other materials: RuO_2, NiCoFeP/NF, and MoP/NF. It was concluded that the NiCoFeMoP/NF catalyst exhibited the highest OER electrocatalytic activity. Its OER activity was better than that of cobalt-based or nickel-based phosphides reported in the literature, as shown in Table 2.

The magnitude of the Tafel slope can also indicate the fast and slow reaction kinetics of the catalytic electrodes in the OER. We determined the Tafel slopes of the catalytic electrodes by analyzing the transformation of the IR-compensated LSV curves to assess the reaction kinetics in OER. The NiCoFeMoP/NF catalyst exhibited a minor slope of 42 mV dec^{-1} among the four catalysts (Fig. 5c). This result indicates that the NiCoFeMoP/NF catalytic electrode exhibits faster reaction kinetics during the OER process. In addition, the Tafel slope and overpotential results obtained from the LSV curves demonstrate the excellent catalytic performance of the NiCoFeMoP/NF material.

Here, to further investigate the OER activity of RuO_2, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF, EIS tests were conducted at 0.46 V (vs. SCE) (Fig. 5d). Comparing the electrochemical impedances of the five samples, the Rct of nickel foam is the highest, followed by the MoP/NF catalyst. This indicates that nickel foam exhibits the slowest charge transfer efficiency and the poorest oxygen evolution performance in the oxygen evolution reaction. Moreover, the Rct of the material decreases and improves when the MoP catalyst is loaded onto the nickel foam. From the detailed magnified plot of electrochemical impedance (Fig. 5e), the NiCoFeMoP/NF catalyst exhibits the smallest Rct and the highest charge transfer efficiency. The high-frequency intersection of the curve with the X-axis in the electrochemical impedance spectrum represents Rs, the resistance primarily caused by the electrolyte and the exposed catalyst substance. At the same time, the diameter of the semicircle indicates Rct. To separate the ohmic resistance from the polarization resistance, we utilize a model where Rs is in series with the module, and Rct is connected in parallel with CPE. The circuit fitting above yields Rcts (Fig. 5d and e) of 1.887, 0.538, 0.729, 20.68, and 27.78 Ω for RuO_2, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF, respectively. The NiCoFeMoP/NF catalyst has the smallest Rct of 0.538 Ω, indicating excellent electrical conductivity and electrocatalytic performance in the alkaline oxygen evolution reaction. This catalyst significantly enhances the charge transfer rate in the oxygen evolution reaction.

Fig. 6 shows the CV plots of four samples: NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF, with different sweep speeds. The corresponding

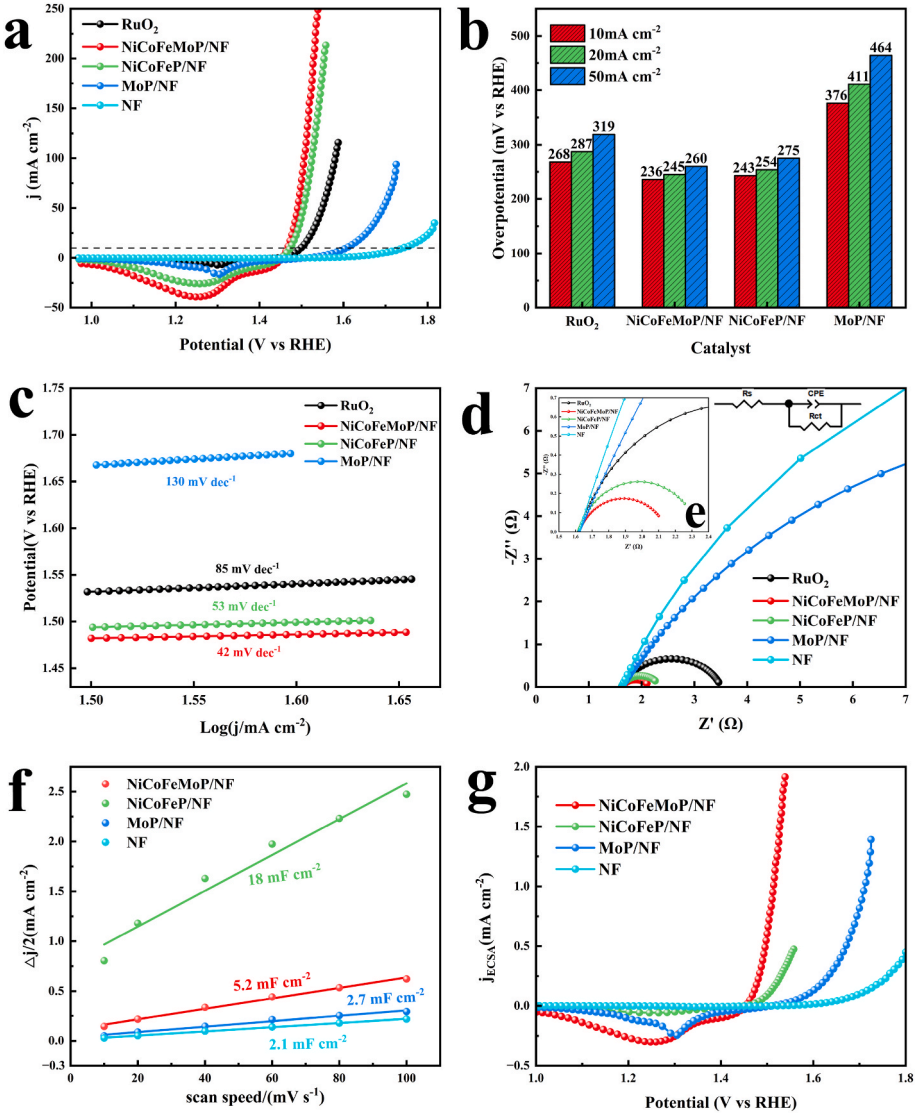


Fig. 5. Oxygen evolution reactions of RuO_2, NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF and NF materials with (a) LSV curves, (b) columnar comparison of overpotentials at different current densities, (c) Tafel plots, (d) electrochemical impedance spectra, (e) magnification of the details of the electrochemical impedance spectra, (f) Cdl plots, (g) ECSA normalized LSV curve.

Table 2
The comparison of the OER performance of the samples.

Sample	Overpotential (mV)	Stability	Electrolyte	Ref.
NiCoP/rGO	$\eta_{10} = 270$	18h	1 M KOH	[61]
Ni_2P–Fe_2P/NF	$\eta_{100} = 261$	48h	1 M KOH	[62]
CoNiFeOx-NC	$\eta_{10} = 265$	40h	1 M KOH	[63]
Co–Mo_2C-$CN_{x\text{-}2}$	$\eta_{10} = 338$	20h	1 M KOH	[64]
$FeNi_2S_4$ NRs/CB	$\eta_{10} = 290$	16h	1 M KOH	[65]
NiCoP NR@NS	$\eta_{10} = 350$	20h	1 M KOH	[66]
$NiCo_2S_4$@NiFe LDH	$\eta_{10} = 287$	10h	1 M KOH	[67]
$NiCo_2O_4$/NiCoP	$\eta_{10} = 295$	10h	1 M KOH	[68]
L-Co_3O_4 NSs	$\eta_{100} = 370$	20h	1 M KOH	[69]
CoP/Mo_2CTx	$\eta_{10} = 260$	50h	1 M KOH	[70]
NiCoFeMoP	$\eta_{10} = 236$	100h	1 M KOH	This work

double-layer capacitance (Cdl) plots of the samples were obtained by extracting points from the CV plots, calculating, and fitting the data (Fig. 5f).

The order of the Cdl values is as follows: NiCoFeP/NF > NiCoFeMoP/NF > MoP/NF > NF. It can be observed that the NiCoFeP/NF catalyst has the highest Cdl value of 18 mF cm^{-2} and a larger ECSA, while the NiCoFeMoP/NF catalyst has the second highest Cdl value of 5.2 mF cm^{-2}. Since the NiCoFeP/NF has a larger electrochemical surface area (ECSA) than the NiCoFeMoP/NF, the LSV curves of the NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF electrodes were then electrochemically normalized to the active area (Fig. 5g). The Cdl was converted to the ECSA using the equation ECSA = Cdl/Cs, where Cs represents the specific capacitance of smooth surfaces, typically assumed to be 40 μF cm^{-2} [71]. The ECSAs of NiCoFeMoP/NF, NiCoFeP/NF, MoP/NF, and NF were calculated to be 130, 450, 67.5, and 52.5 cm^2, respectively. The ECSA normalization of the LSV curves indicated that the oxygen evolution performance of NiCoFeMoP/NF is superior to that of NiCoFeP/NF catalysts. This is evident despite the electrochemically active surface area of the NiCoFeP/NF catalyst being much larger than that of the NiCoFeMoP/NF catalyst. Therefore, the NiCoFeMoP/NF catalyst still exhibits the best oxygen evolution performance.

3.4. HER/OER stability test

The above tests of HER and OER demonstrated the superior performance of the NiCoFeMoP/NF catalytic electrode. To investigate the HER and OER stability, the hydrogen/oxygen evolution of NiCoFeMoP/NF

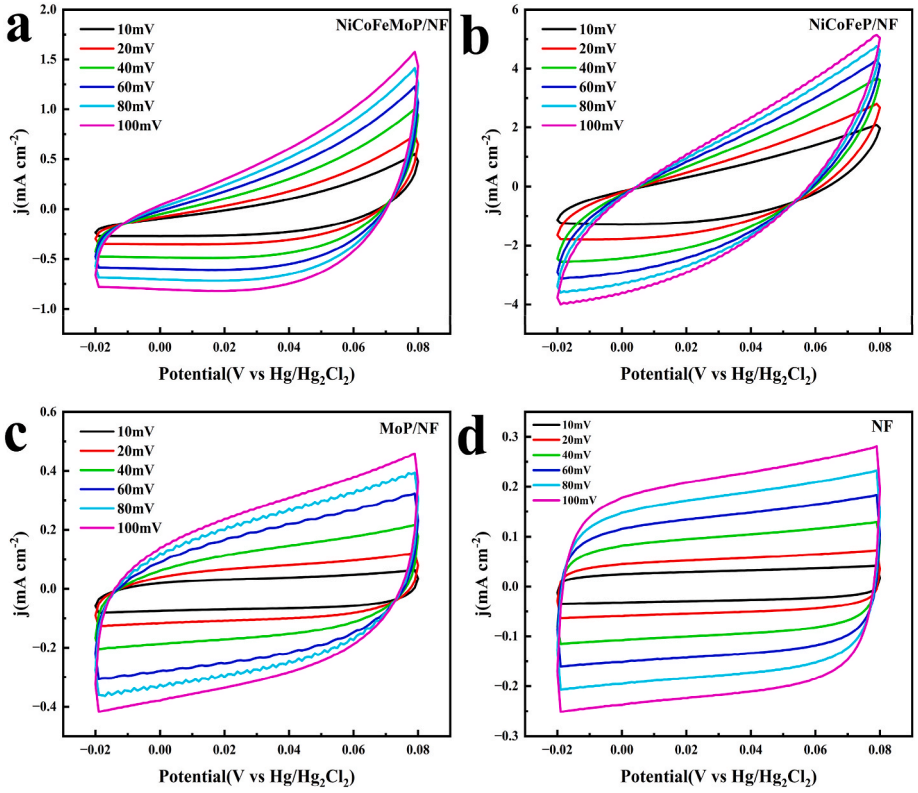


Fig. 6. CV plots for different sweep speeds: (a) NiCoFeMoP/NF, (b) NiCoFeP/NF, (c) MoP/NF, (d) NF.

catalytic electrode over an extended period at a constant current density of 10 mA cm^{-2} was tested (Fig. 7).

Fig. 7a shows the stability tests of the hydrogen evolution reaction of the NiCoFeMoP/NF catalytic electrode. The voltage of the catalyst increased slightly but remained relatively stable , suggesting that the NiCoFeMoP/NF catalyst exhibits excellent long-term stability. Fig. 7b shows the step-current operation test of the NiCoFeMoP/NF catalytic electrode. The potential change at different current densities remains

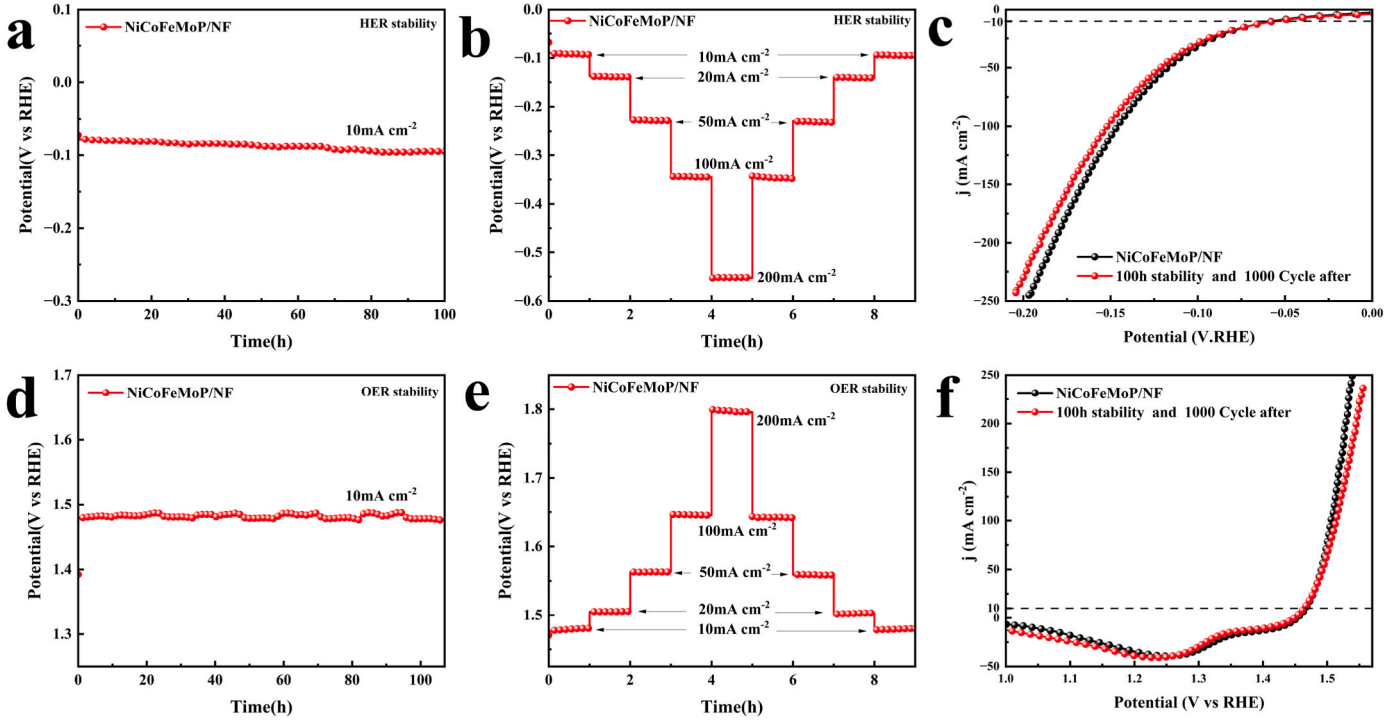


Fig. 7. Hydrogen evolution reaction of NiCoFeMoP/NF catalyst (a) long time stability test, (b) step current running test, (c) LSV curve after 100 h stability test and 1000 cycles CV test; Oxygen evolution reaction of NiCoFeMoP/NF catalyst (d) long time stability test, (e) step current running test, (f) 100 h stability test and LSV curves after 1000-lap CV test.

stable during the hydrogen evolution reactions at a step-current density of 10–200 mA cm^{-2}. This stability indicates that the NiCoFeMoP/NF catalytic electrode exhibits excellent stability at low or high current densities. In Fig. 7c, the black LSV curves represent the initial test. In contrast, the red LSV curves depict the NiCoFeMoP/NF catalysts after the 100-h stability test and the 1000-turn CV test at a constant current density of 10 mA cm^{-2}. Compared to the initial state, the LSV curves from the long-time stability test and the multi-turn CV test exhibit more excellent stability at low current densities. The LSV curves remain primarily unchanged at low current densities, with only a slight increase in overpotential observed at high current densities.

Fig. 7d–f shows the stability tests of the oxygen evolution reaction of the NiCoFeMoP/NF catalytic electrode. As shown in Fig. 7d, the NiCoFeMoP/NF catalytic electrode was tested for oxygen reduction for an extended period at a constant current density of 10 mA cm^{-2} without any significant change in voltage, indicating excellent long-term stability. Fig. 7e shows the step-current operation test of the NiCoFeMoP/NF catalytic electrode. The potential change at different current densities remains stable at a step-current density of 10–200 mA cm^{-2} during the oxygen evolution. This suggests that the NiCoFeMoP/NF catalytic electrode exhibits excellent stability at low or high current densities. In Fig. 7f, the black polarization curves represent the initial LSV curves of NiCoFeMoP/NF catalysts, while the red LSV curves depict the LSV curves of NiCoFeMoP/NF catalysts after more than 100 h of stability testing and 1000 cycles of CV testing at a constant current density of 10 mA cm^{-2}, in comparison to the initial LSV curves. After the long-term stability and multi-turn CV tests, the polarization curves show no change in overpotential at low current density and only a slight increase in overpotential at high current density. From the stability test, it is concluded that the excellent stability of the NiCoFeMoP/NF electrode is crucial for its practical application in hydrogen and oxygen evolution reactions.

3.5. OWS performance

To investigate the water electrolysis performance, the two-electrode system was used for the catalytic electrode measurement. The LSV diagram is shown in Fig. 8a. The water electrolysis device consists of two identical NiCoFeMoP/NF electrodes and achieves a current density of 10 mA cm^{-2} at a voltage of only 1.558 V. To further assess the stability of the catalytic electrodes for complete water splitting, stability tests were conducted at a current density of 10 mA cm^{-2} for over 100 h for catalytic systems: NiCoFeMoP/NF and NiCoFeP/NF (Fig. 8b). It was observed that the voltage for total water splitting in the NiCoFeMoP/NF catalytic system remained essentially unchanged, while the NiCoFeP/NF catalytic system exhibited an increase in the voltage. Secondly, the potentials of the NiCoFeMoP/NF catalytic system remain stable at different current densities throughout the overall water-splitting process under a step current density of 10–200 mA cm^{-2} (Fig. 8c), which exhibits excellent stability, whether operating at low or high current densities.

As shown in Fig. 8d, the LSV curves were tested after the 100-h ultra-stability and 1000-lap CV tests. Compared with the initial LSV curves (Fig. 8a), the LSV curves after the long-term stability test and the multi-lap CV test showed no change in the total desalination voltage and only a slight increase in high current density. It can be seen that the NiCoFeMoP/NF catalytic electrode exhibits excellent performance in total water dissociation and stability.

3.6. DFT calculation

To study the effects of Fe and Mo elemental doping on the electrocatalytic activity of NiCoP, the water dissociation barriers (ΔG_B), hydrogen adsorption Gibbs free energies (ΔG_{H^*}), and densities of states (DOS) of NiCoP, NiCoFeP, and NiCoFeMoP were computed and analyzed in this paper by DFT calculations (Fig. 9). Additionally, OER step diagrams were plotted for further analysis.

The HER process in alkaline media primarily involves the adsorption and dissociation of water on the catalyst, forming *H intermediates and

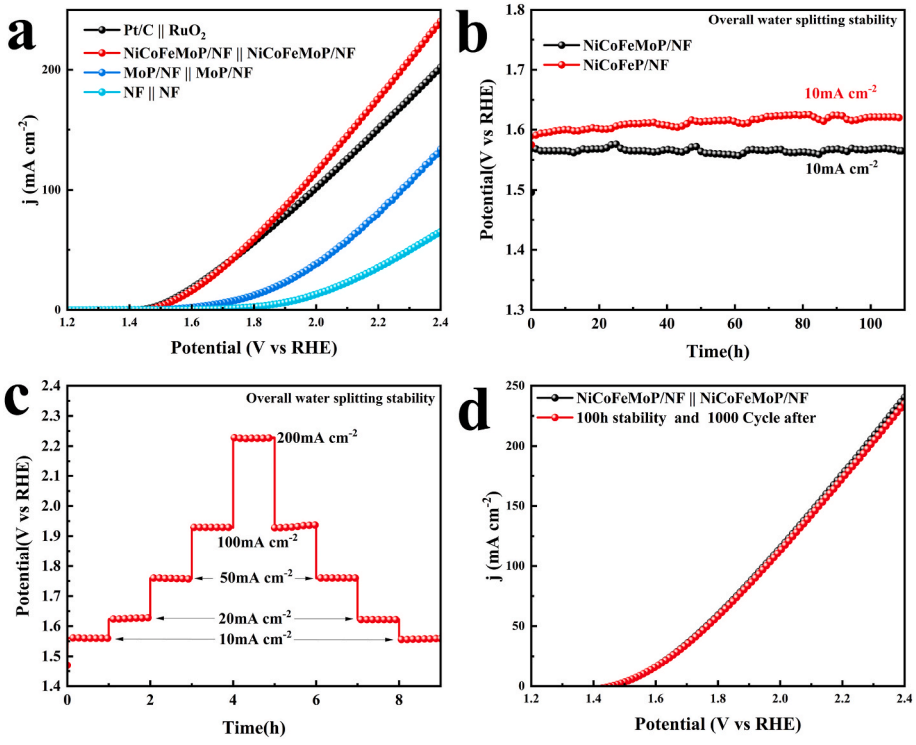


Fig. 8. LSV curves for the total solution water reaction test (a) for RuO_2, NiCoFeMoP/NF, MoP/NF and NF, (b) stability curves for NiCoFeMoP/NF and NiCoFeP/NF catalysts, (c) step-current operating curves for NiCoFeMoP/NF catalysts, (d) NiCoFeMoP/NF LSV curves after 100 h stability test and 1000 turns CV test of catalysts.

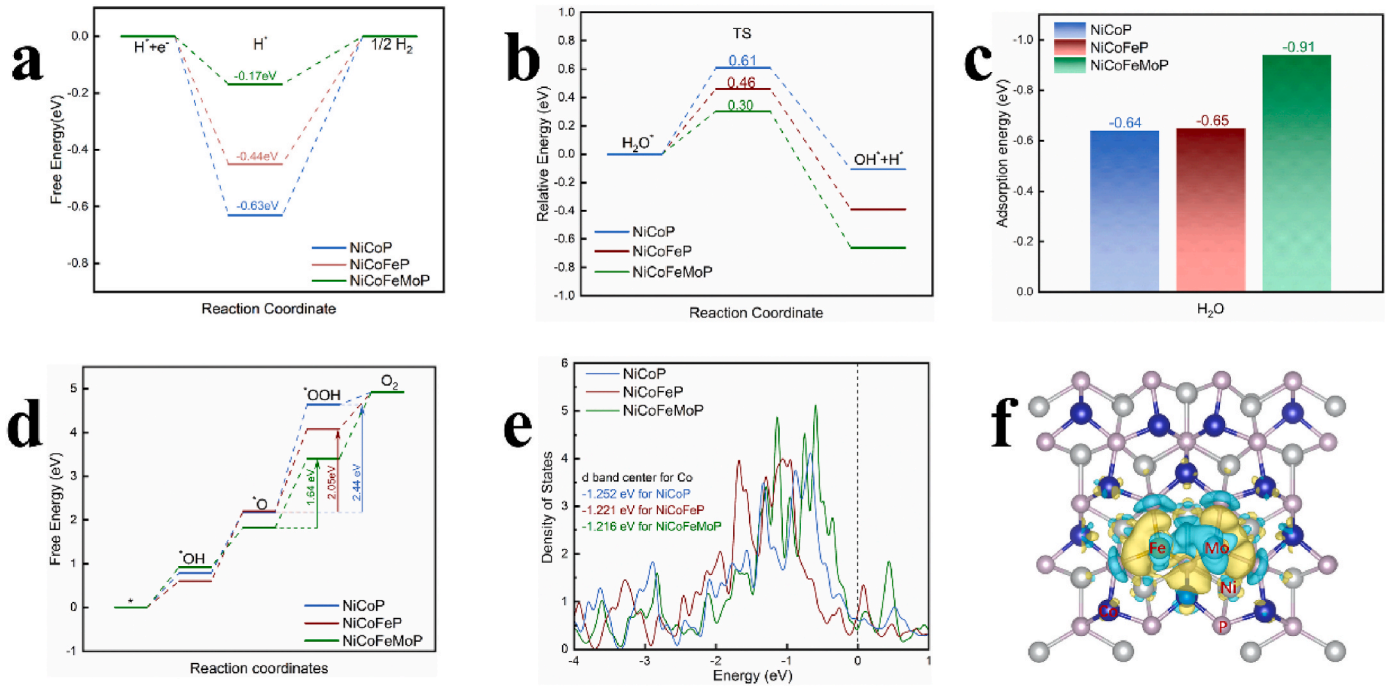


Fig. 9. Hydrogen adsorption Gibbs free energy diagrams (a), water dissociation free energy diagrams (b), water molecule adsorption energies (c), OER step diagrams (d), the density of states diagrams (e) for NiCoP, NiCoFeP, NiCoFeMoP, and charge differential density diagrams (f) for NiCoFeMoP.

generating H_2 [72]. The free energy of hydrogen adsorption (ΔG_{H^*}) is commonly used to characterize the HER activity. Catalysts with ΔG_{H^*} values close to zero are considered the best candidates for catalyzing the HER process because they can facilitate hydrogen release and proton/electron transfer in a much shorter time [44,73]. Fig. 9a and Fig. S2 shows the Gibbs free energy plots and configuration of hydrogen adsorption for three materials: NiCoP, NiCoFeP, and NiCoFeMoP, with ΔGH^* of −0.63 eV, −0.44 eV, and −0.17 eV, respectively. Among them, the ΔGH^* of NiCoFeMoP is the closest to 0, suggesting it has a more favorable hydrogen adsorption strength. Moreover, in an alkaline electrolytic water environment, the kinetics of HER are mainly limited by its slow Volmer step, which involves the dissociation of water molecules into active hydrogen [74]. In this paper, the free energies of water dissociation on NiCoP, NiCoFeP, and NiCoFeMoP are calculated, as shown in Fig. 9b and Fig. S3. NiCoP exhibits a largest hydrolysis dissociation barrier ($\Delta G_B = 0.61$ eV), indicating the difficulty for water molecules to dissociate on its surface to produce active hydrogen. Additionally, its Volmer step is slow. After Fe doping, the hydrolysis barrier decreased to 0.46 eV, and the Volmer step was enhanced. After co-doping with Fe and Mo elements, the hydrolysis dissociation barrier was significantly reduced to 0.30 eV, and its Volmer step was greatly accelerated [75]. In addition, according to the water molecule adsorption energy diagram shown in Fig. 9c, NiCoFeMoP has the largest adsorption energy for water molecules (−0.91 eV). This suggests that water molecules are more likely to be adsorbed on NiCoFeMoP surface than NiCoP and NiCoFeP. This ensures that a sufficiently large number of water molecules can be present on the surface for decomposition for the HER/OER reaction [76]. To investigate the effect of Fe and Mo doping on the electronic properties of NiCoP, the density of states of NiCoP, NiCoFeP, and NiCoFeMoP was calculated, as shown in Fig. 9e. The density of states near the Fermi energy level of the three materials increases gradually, indicating that the electronic properties of NiCoP are enhanced with the doping of Fe and Mo elements. This enhancement allows for faster electron transfer, as supported by the improved electronic properties of NiCoP due to the doping of Fe and Mo elements, enabling the reaction of HER/OER. More rapid electron transfer is also consistent with the results of the electrochemical impedance test. Meanwhile, by observing the charge density difference shown in Fig. 9f (where the yellow area indicates electron aggregation and the blue area indicates electron dissipation, with an isosurface of 0.0035 e•a^{-3}), it is evident that the Fe and Mo atoms lose their electrons after doping, and form the formation of high valence states of Fe^{2+}, Fe^{3+}, and Mo^{4+}, aligning with the findings of the XPS analyses.

According to the OER four-electron reaction mechanism [77–79], in this paper, ΔG_{OH^*}, ΔG_{O^*}, and ΔG_{OOH^*} were calculated for three materials, NiCoP, NiCoFeP, and NiCoFeMoP (Fig. 9d), where the largest step of the free energy change is the rate-determining step (RDS) of the OER. From the results, the maximum potential of NiCoP, NiCoFeP, and NiCoFeMoP is the formation of OOH* ($H_2O + O^* \rightarrow OOH^* + H^+ + e^-$), which is its rate-determining step, with free energy changes of 2.45 eV, 2.05 eV, and 1.64 eV, respectively. It is evident from the comparisons that the lowest free energy change of the rate-determining step of NiCoFeMoP indicates the best catalytic performance, aligning with the experimentally obtained results.

We also calculated the d-band centers of the active sites of NiCoP, NiCoFeP, and NiCoFeMoP. The d-band center of the Co active site in NiCoP is −1.252 eV, which slightly increases to −1.221 eV after doping with Fe, and further increases to −1.216 eV after doping with Fe and Mo. According to the d-band center theory proposed by Hammer and Nørskov [71,80], an upward shift of the d-band center results in a reduction in the occupancy of the antibonding state between the active site and the intermediate product. This shift enhances the adsorption of water molecules and the intermediate product *OOH on the catalyst, thereby facilitating the optimization of the electrocatalytic activity. This is corroborated by its water molecule adsorption energy and OER step diagrams. It also demonstrates that the doping of Fe and Mo elements modulates the electronic structure of NiCoP, optimizing the adsorption energy of the reaction intermediates, which leads to excellent OER performance.

The adsorption active sites of NiCoFeMoP involving HER and OER sub-steps are shown in Fig. 10.

It can be observed that H_2O first adsorbs and then dissociates on the Fe–Mo bridge sites to form *H and *OH. Subsequently, the *H atoms and the H_2O molecules adsorbed on NiCoFeMoP produce hydrogen gas,

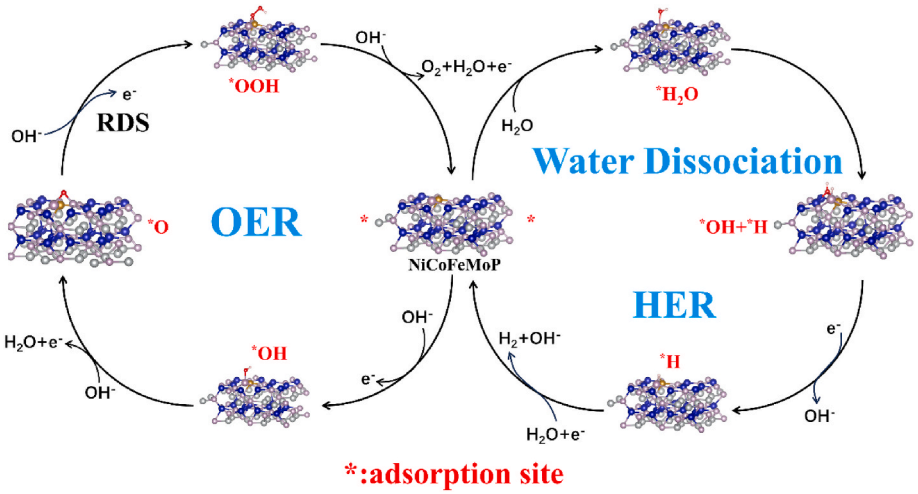


Fig. 10. Intrinsic mechanism of OER/HER reaction over NiCoFeMoP catalysts.

completing the HER process. Then NiCoFeMoP adsorbs OH^-, which leads to the conversion of *OH to *O, completing the process. After this step, *O adsorbs OH^- to form *OOH. Finally, the OER process ends with the release of oxygen and the reduction of adsorption sites.

4. Conclusions

In this paper, NiCoFeMoP/NF self-supported bifunctional electrocatalysts with petal clusters were prepared using a strategy that combines a one-step hydrothermal method and a low-temperature phosphating method. The unique nanostructures, abundant catalytic sites, and the doping effects of Fe and Mo elements have endowed NiCoFeMoP/N bifunctional catalysts with excellent catalytic activities. In the hydrogen evolution reaction, the overpotential of the NiCoFeMoP/NF catalyst at a current density of -10 mA cm^{-2} was 58 mV. In the oxygen evolution reaction, the overpotential of the NiCoFeMoP/NF catalyst at a current density of 10 mA cm^{-2} was 236 mV, showcasing the superior electrocatalytic activity for both HER and OER. It was superior to the Pt electrode when the current exceeded -100 mA cm^{-2}. The NiCoFeMoP/NF catalyst requires only 1.558 V to achieve a current density of 10 mA cm^{-2} in the overall water-splitting process and exhibits excellent stability. Based on experiments and DFT calculations, the electronic structure of the NiCoP active site can be modified by introducing Fe and Mo elements to optimize the adsorption energy of hydrogen and oxygen-containing intermediates, thereby facilitating the HER/OER process. This study demonstrates a new approach to designing efficient electrocatalysts using heteroatom doping and structural engineering strategies, which can enhance the efficiency of energy conversion technologies.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Jianfeng Wan: Validation, Project administration, Methodology, Formal analysis, Conceptualization. **Yihong Sun:** Writing – original draft, Software, Investigation, Data curation. **Qiang Zhou:** Formal analysis, Data curation. **Wenyan Bi:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis. **Shizheng Xie:** Data curation. **Yikai Hou:** Software. **Menglin Yu:** Software. **Tianen Li:** Resources, Conceptualization. **Baozhong Liu:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2024.09.338.

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